SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA03	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	**Total Marks:**	50 (5 Questions × 10 Mark)
CO5 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity (BL3, BL4)		
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Section / Batch:	2024-A	**Date:**	16-7-2026

Code of Conduct

I K .Deeksha Sree_____________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K .Deeksha Sree____________

Instructions

- This test contains 5 Application-Based Questions Total: 50 Marks.
- Evaluation of Answer Quality - Concept Identification, Linkages, Organization, Integration of Literature and clarity
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Introduction to Data Structure- Abstract Data Types (ADT)	ADT	1,2	20
Types of Data Structure- Linear and Non linear Data structures	Linear and Non linear Data structures	3,4	20
Recursion , Performance Analysis and Measurement	Performance Analysis	5	10
		GRAND TOTAL:	**50Marks**

Annexure A – Concept Map Questions

Q1. A system needs to search for specific records in a large dataset efficiently. Construct a concept map linking Searching Algorithm → Linear Search → Binary Search → Time Complexity → Efficiency. Include relationships that compare the number of operations and performance of each method. Analyze which algorithm is more efficient and justify your conclusion.

Q2. A library management system stores thousands of book records including title, author, and availability status. Construct a concept map linking Data → Data Structure → Organization → Access Methods → Efficiency. Include different data structures used for storing and retrieving records. Explain how proper organization improves search efficiency and system performance

Q3. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing..

Q4. A logistics company processes delivery requests across multiple cities. Construct a concept map linking Algorithm → Input Size → Time Complexity → Space Complexity → Scalability → Efficiency. Analyze how algorithm selection impacts system scalability and justify the best approach for handling large datasets.

Q5. A printer system manages print jobs submitted by multiple users. Construct a concept map connecting Queue → FIFO → Job Scheduling → Processing → Efficiency and Stack → LIFO → Undo Operations → Execution. Explain how each structure is applied and justify their suitability in this context.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Faculty In-charge	Course Coordinator	Program Director
Dr.S.Kalaiarasi		
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Searching Algorithm Concept Map

Searching Algorithm

Linear Search → Checks every element one by one → O(n) Time → Less efficient

Binary Search → Divides sorted array into halves repeatedly. → O(log n) Time → More efficient

Searching algorithm used to find a required element from a collection of data.

Linear search checks each element one by one until the target is found.

Binary search repeatedly divides the sorted list into two halves to locate the target.

Binary search is more efficient because it reduces the search space by half after every comparison. For large data it performs far fewer operation than linear search. Binary search requires data to be sorted.

2. Library Management System Concept Map

Data

↓

Stored using

↓

Data structure

Array   Linked List   Hash Table

Organisation

Access Methods
(Search, Insert, Delete)

Efficiency

Data includes book title, author, ISBN and availability
Data structure organises, data for easy storage.
Arrays store records sequentially.
Linked lists allow easy insertion and deletion
Hashtables provide very fast searching
Proper organisation improves quick access and reduces search time.
Data structures improves search speed, reduces, processing time and increases overall system performance.

3. Weather Monitoring System Concept Map

Primitive Data types
(int, float, char)

Used to build

Non-Primitive structures

Linear structures   Non linear structures

Array, Queue   Tree, Graph

Primitive Data types store temperature, humidity and wind speed values.
Non-Primitive structures organise multiple data values
Linear structures (Arrays, Queues) process data in sequence.
Non-Linear structures (Trees, Graphs) manage complex relationship
Structures support continuous weather data collection, storage and analysis.

4. Algorithm Performance Concept Map

Algorithm
↓
Input Size
↓
Time Complexity
↓
Space Complexity
↓
Scalability
↓
Efficiency

An algorithm solves a problem. Input size affects execution time. Time complexity measures how long the algorithm takes. Space complexity measures memory usage. Good algorithms improve scalability, allowing systems to handle larger datasets efficiently.
Algorithms with lower time complexity (such as $O(\log n)$ or $O(n \log n)$) are more suitable for large datasets because they execute faster and scale better than algorithms with $O(n^2)$ complexity.

5. # Queue and Stack Concept Map

Queue

FIFO
(First in First out)

Job Scheduling

Processing

Efficiency

Stack

LIFO
(last in first out)

Undo operations

Execution

Queue follows FIFO (First in, First out). The first print job submitted is printed first.

Used in printer job scheduling

Stack follows LIFO (last in first out). Most recent action is undone first. Used in undo operations in text editors.

Queue is best for printer systems because jobs are processed in the order they arrive.

Stack is best for undo operations because the latest action should be reserved first.